

STOCK PRICE FORECASTING: LSTM WITH BAHDANAU ATTENTION

Equity Market Time Series Forecasting and Sequence Window Analysis

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Domain: Deep Learning Time Series

Stack: PyTorch, TensorFlow, Streamlit

PERFORMANCE METRICS

2.3%

Test Set RMSE

18.0%

vs Baseline Boost

60 Days

Sequence Window

5+ Years

OHLCV Market Data

ARCHITECTURE AND INSIGHTS

Engineered for NSE and BSE equities, this forecasting pipeline integrates a Bahdanau attention mechanism into a Long Short Term Memory architecture. By dynamically weighting historical time steps across a 60 day window, the network highlights critical market inflection points while filtering out noise.

- Data Source: Yahoo Finance API ingesting 1,250 daily OHLCV trading bars
- Local Data Cache: Stored in data/nifty50_stock_data.csv
- Error Reduction: Achieved 2.3% RMSE on NIFTY 50 equities, outperforming vanilla LSTM models by 18.0%